A8 (25 marks)

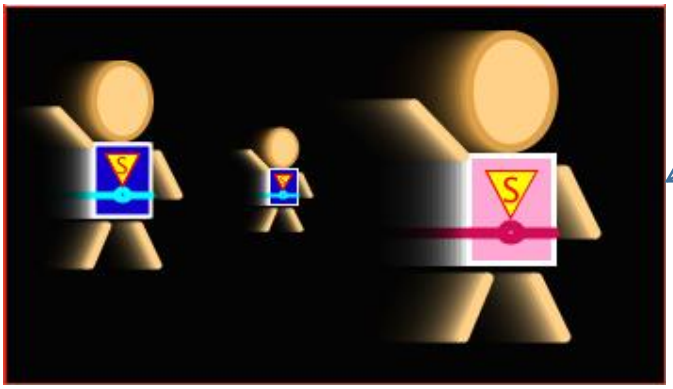
Focus: *OOP*

For each question in this assignment, you need to do the following:

1) Use the starter code.
2) Fill in the missing code as per the instructions given in the starter code. These instructions are written as comments which start with "REQ".

Q1. [10] In this question, you will create a class named `Hero` for your custom character (the one that we created in the beginning of this semester). Here are the specs for the class:

- **Attributes**: `x`, `y`, `scale`, `speedX`, `speedY`, `torsoColor`, and `beltColor`.
- **Constructors**:
  - zero-argument constructor that sets attributes as: `x=width/2`, `y=height/2`, `scale=1`, `speedX=1`, `speedY=0`, `torsoColor=color(19,0,205)`, and `beltColor=color(0,255,255)`.
  - 3-argument constructor to set `x`, `y` and `scale` to given values. Other attributes should use the same values used in the zero argument constructor. Use this header: `Hero(float x1, float y1, float scale1)`
  - 7-arg constructor to set all attributes to given values. Use this header: `Hero`(float x1, float y1, float scale1, float sx, float sy, color torsoClr, color beltClr)
- **Functions**:
  - `move()` to move the character and *wraps* it around the screen when going outside sketch boundaries (e.g. when it passes the right edge, it should reappear at the left edge).
  - `display()` to draw the character as per the design that you made before.



Your character will probably have a different design than mine!

Marking Guide

+2 for Hero class header and attributes
+3 for three constructors.
+3 for move function (with wrapping).
+2 for display function.

Q2. [15] In this question, you will rewrite you solution from the previous assignment (A7) but using OOP. You will need to create a class `Button` for your buttons (create other classes if you wish, but the marking will be based on the `Button` class only). A `Button` class should have:

- Attributes for the width, height, position (x,y), caption (text), and fill color.
- Two constructors:
  - A zero-argument constructor to set the above attributes to default values.
  - A six-argument constructor to set the above attributes to given values.
- Functions:
  - `boolean mouseOver()` that returns `true` if the mouse is over the button, and `false` otherwise.
  - `void display()` to draw a button object. This should be the same as the `drawButton()` function from the starter-code of assignment A7 except that `display()` should have no arguments. Instead, it should use the above attributes to determine how the button will be displayed. Note that the button's outline should always be white.

Marking Guide

+2	for creating two buttons
+1	for drawing either scene0 or scene1 (under the draw() function).
+8	for Button class (+1 for attributes, +2 for constructors, +3 for mouseOver, +2 for display)
+2	for the scene0 (+1 for copied code, +1 for calling button b0's display function).
+2	for the scene1 (+1 for copied code, +1 for calling button b1's display function).

Submission Instructions

For this assignment, you need to do the following:

1- Copy your solution into this Word document template.
2- Compress your solution document along with the images you used into one zip file.
3- Submit your zip file to Canvas. ***Both students in a team must submit the same file***.

Note that you can resubmit an assignment, but the new submission overwrites the old submission and receives a new timestamp.